

CHAPTER 1 Introduction

Computer science deals with the theoretical foundations of information, its computation and practical techniques for their applications.

Peter Denning: the fundamental question underlying computer science is

What can be automated?

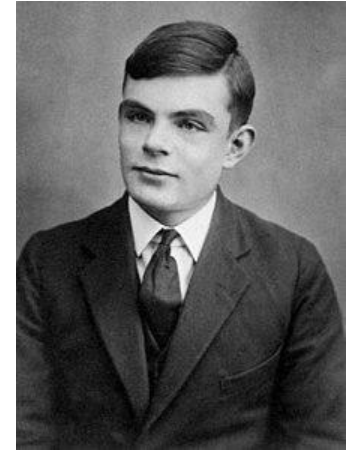
Objectives

After studying this chapter, the student should be able to:

- Define the Turing model of a computer.
- Define the von Neumann model of a computer.
- Describe the three components of a computer: hardware, data, and software.
- List topics related to computer hardware.
- List topics related to data.
- List topics related to software.
- Give a short history of computers.

1.1 TURING MODEL

The idea of a universal computational device was first described by Alan Turing in 1937.

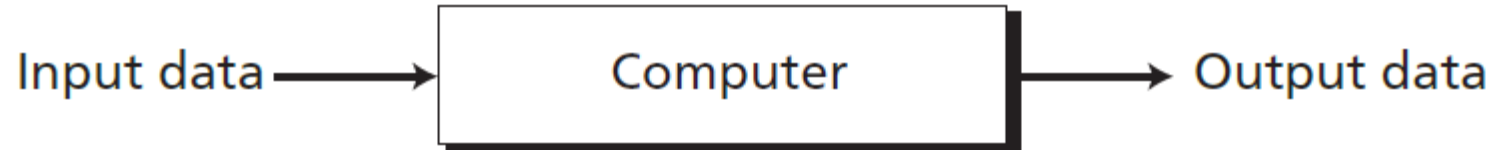


- He proposed that all computation could be performed by a special kind of a machine, now called a **Turing machine**.
- He based the model on **the actions that people perform** when involved in computation.
- He abstracted these actions into a model for a computational machine that has really changed the world.

1.1.1 Data processors

- Before discussing the Turing model, let us define a computer as a data processor.
- Using this definition, a computer acts as a black box that accepts input data, processes the data, and creates output data (Figure 1.1).

Figure 1.1 *A single-purpose computing machine*



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- Although this model can define the functionality of a computer today, it is too general.
 - In this model, a pocket calculator is also a computer.

1.1.2 Programmable data processors

- The **Turing model** is a better model for a general-purpose computer.
- This model adds an extra element to the specific computing machine: **the program**.
- A program is a set of instructions that tells the computer what to do with data.
- Figure 1.2 shows the Turing model.

Figure 1.2 *A computer based on the Turing model: programmable data processor*

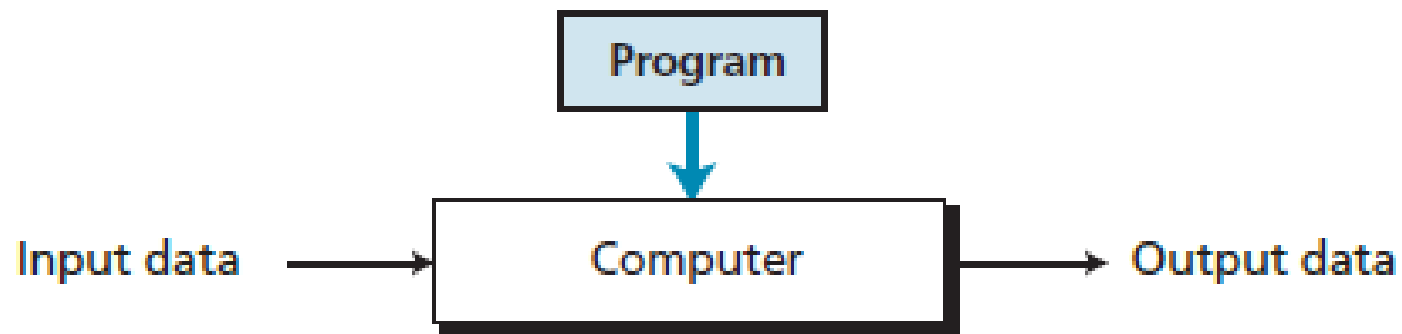


Figure 1.3 *The same program, different data*

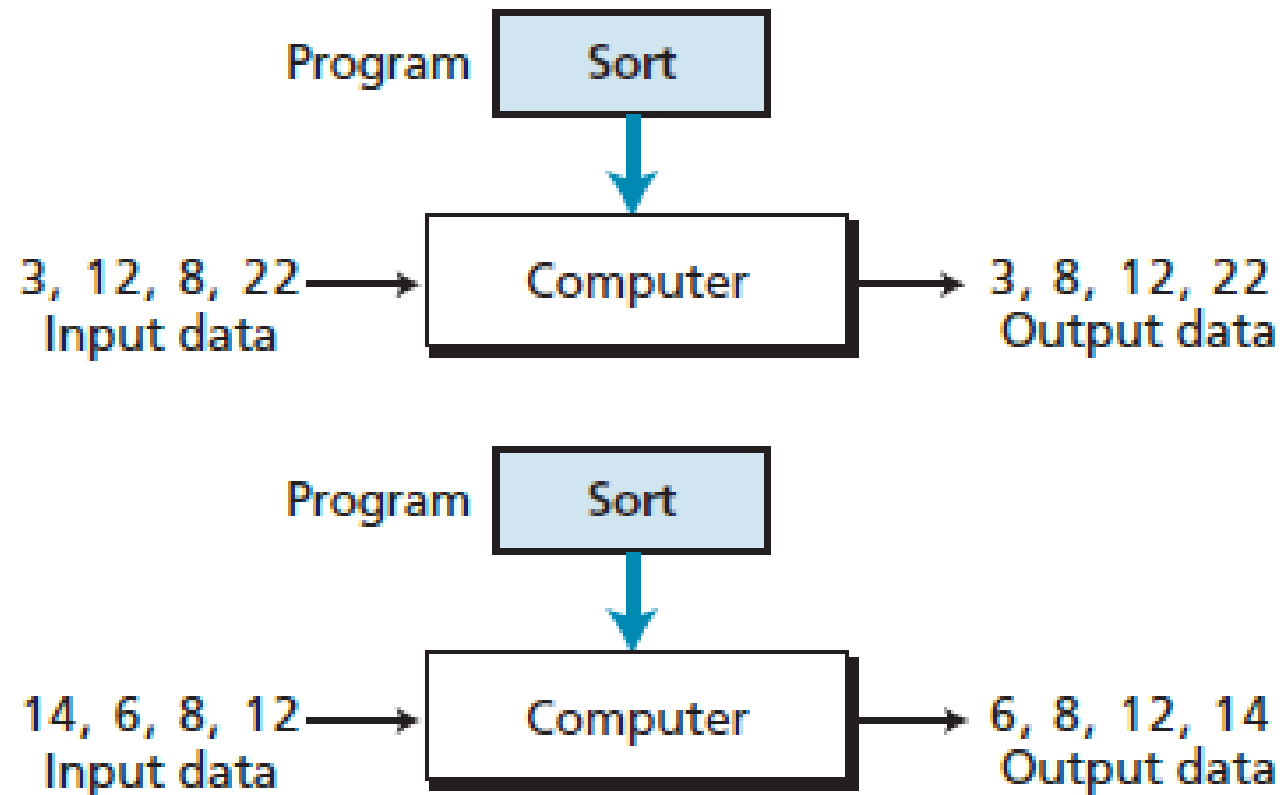
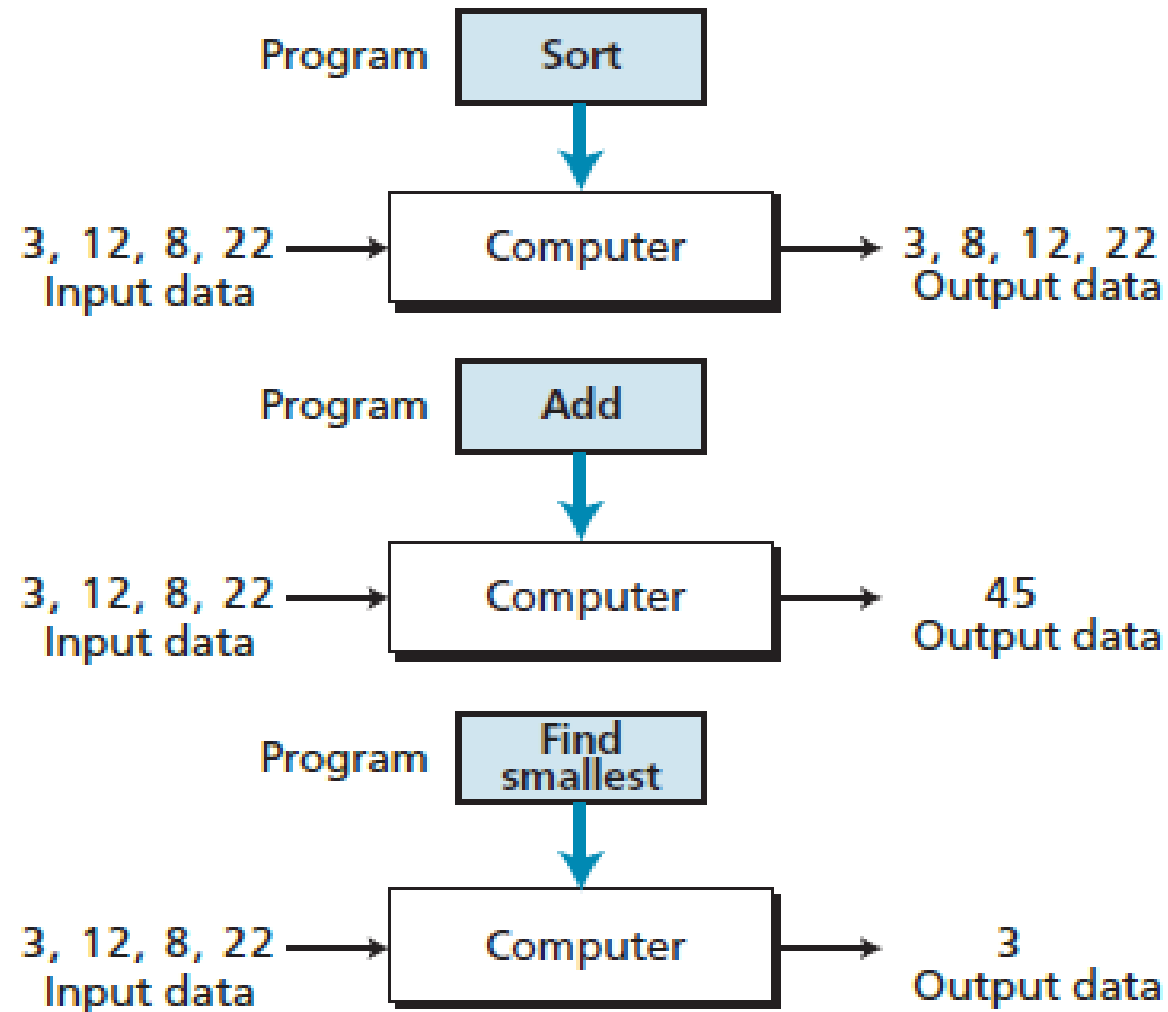


Figure 1.4 *The same data, different programs*



1.1.3 The universal Turing machine

Universal Turing machine, a machine that can do any computation if the appropriate program is provided, was the first description of a modern computer.

- It can be proved that a very powerful computer and a universal Turing machine can compute the same thing.
- We need only provide the data and the program—the description of how to do the computation—to either machine.
- In fact, **a universal Turing machine is capable of computing anything that is computable.**

1.2 VON NEUMANN MODEL

- Computers built on the Turing universal machine store data in their memory.
- Around 1944–1945, John von Neumann proposed that, **since program and data are logically the same, programs should also be stored in the memory of a computer.**

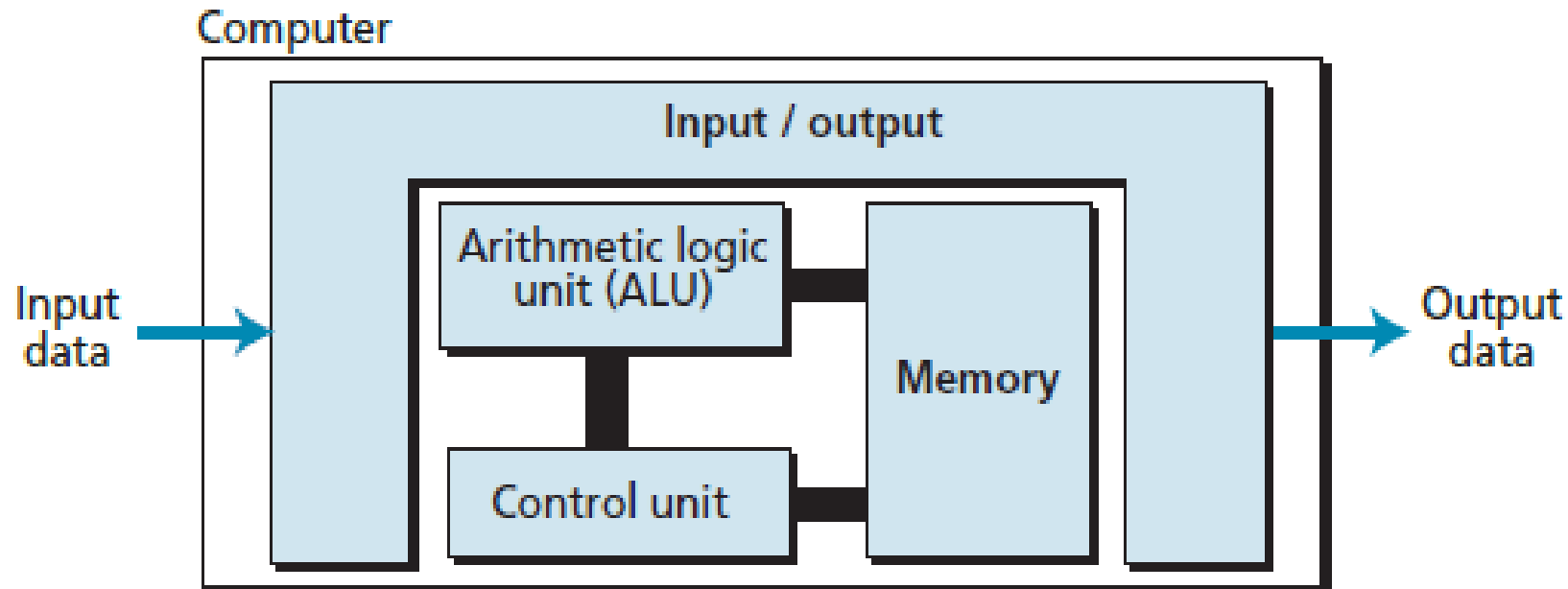
The von Neumann architecture—also known as the von Neumann model or Princeton architecture—is a computer architecture based on a 1945 description by Hungarian-American mathematician and physicist John von Neumann and others in the *First Draft of a Report on the EDVAC*.



1.2.1 Four subsystems

- Computers built on the **von Neumann model** divide the computer hardware into four subsystems: **memory**, **arithmetic logic unit**, **control unit**, and **input/output** (Figure 1.5).

Figure 1.5 *The Von Neumann model*



1.2.2 The stored program concept

- The von Neumann model states that **the program must be stored in memory**.
 - This is totally different from the architecture of early computers in which only the data was stored in memory: the programs for their task was implemented by manipulating a set of switches or by changing the wiring system.
- The memory of modern computers hosts both a program and its corresponding data.
 - This implies that both the data and programs should have the same format, because they are stored in memory.
 - In fact, they are stored as binary patterns in memory—a **sequence of 0s and 1s**.

1.2.3 Sequential execution of instructions

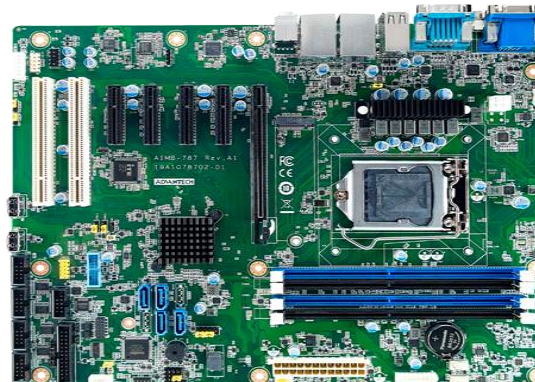
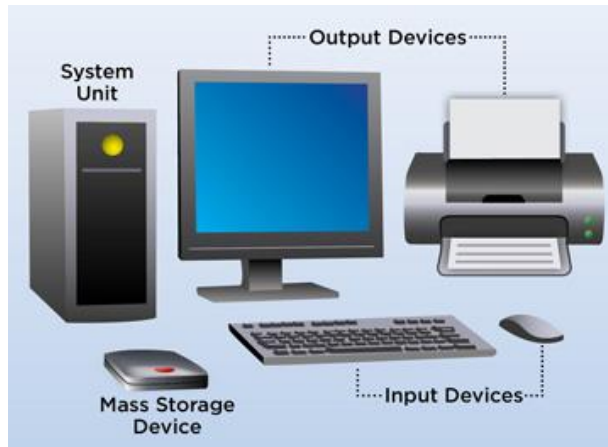
- A program in the von Neumann model is made of a finite number of instructions.
- In this model, the control unit **fetches** one instruction from memory, **decodes** it, then **executes** it. In other words, the instructions are executed one after another. Of course, one instruction may request the control unit to jump to some previous or following instruction, but this does not mean that the instructions are not executed sequentially.
- **Sequential execution of a program** was the initial requirement of a computer based on the **von Neumann model**.
- Today's computers execute programs in the order that is the most efficient.

1.3 COMPUTER COMPONENTS

- We can think of a computer as being made up of three components: **computer hardware**, **data**, and **computer software**.

1.3.1 Computer hardware

- Computer hardware today has four components under the von Neumann model, although we can have different types of memory, different types of input/output subsystems, and so on.



1.3.2 Data

- The von Neumann model clearly defines a computer as a data processing machine that accepts the input data, processes it, and outputs the result.

The von Neumann model clearly defines a computer as a data processing machine that accepts the input data, processes it, and outputs the result.

text



image



sound

1.3.3 Computer software

- The main feature of the Turing or von Neumann models is the concept of the program. Although early computers did not store the program in the computer's memory, they did use the concept of programs.
- Programming those early computers meant changing the wiring systems or turning a set of switches on or off. Programming was therefore a task done by an operator or engineer before the actual data processing began.



Figure 1.6 *Program and data in memory*

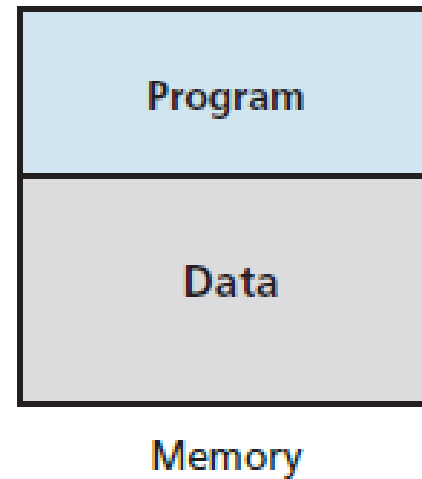


Figure 1.7 *A program made of instructions*

1. Input the first number into memory.
2. Input the second number into memory.
3. Add the two together and store the result in memory.
4. Output the result.

Program

1.4 HISTORY

Timeline of Computer History

In this section we briefly review the history of computing and computers. We divide this history into three periods.

- Mechanical machines (before 1930)
- The birth of electronic computers (1930–1950)
- Computer generations (1950–present)

1.4.1 Mechanical machines (before 1930)

During this period, several computing machines were invented that bear little resemblance to the modern concept of a computer.

- In the 17th century, **Blaise Pascal**, a French mathematician and philosopher, invented **Pascaline**.
 - 一種機械式計算器，利用齒輪和輪盤的轉動來進行 加法與減法的計算。
 - 被認為是世界上最早的機械計算器之一。
- In the late 17th century, German mathematician **Gottfried Leibnitz** invented **Leibnitz' Wheel**.
 - 一種改良自帕斯卡計算器的機械裝置，透過步進鼓 (stepped drum) 的設計，可以進行更複雜(加法、減法、乘法與除法)的計算。
 - 萊布尼茲輪的設計比帕斯卡計算器更先進，不僅擴展了計算功能，也成為後來許多計算機械與現代電腦運算邏輯的基礎。



(Pascal's calculator)



(Leibnitz' Wheel)

- The first machine that used the idea of storage and programming was the **Jacquard loom**, invented by **Joseph-Marie Jacquard** at the beginning of the 19th century.
 - 第一個應用「儲存」與「程式化」概念的機器是傑卡德織布機
 - 使用 穿孔卡片 (punch cards) 來「儲存」織布圖案。
 - 每張卡片就像一個「程式指令」，告訴織布機哪些針要抬起或放下。
 - 透過更換卡片組合，就能「程式化」不同的布料花樣。
- In 1823, **Charles Babbage** invented the **Difference Engine**. 差分機
 - Later, he invented a machine called the **Analytical Engine** that parallels the idea of modern computers. 這台機器的概念非常接近現代電腦。
- In 1890, **Herman Hollerith**, working at the US Census Bureau (美國人口普查局), designed and built a programmer machine that could automatically read, tally, and sort data stored on punched cards.



1.4.2 The birth of electronic computers (1930–1950)

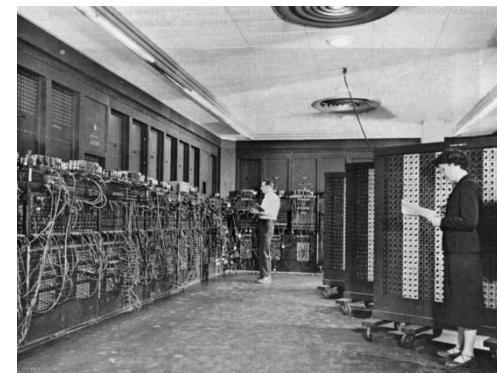
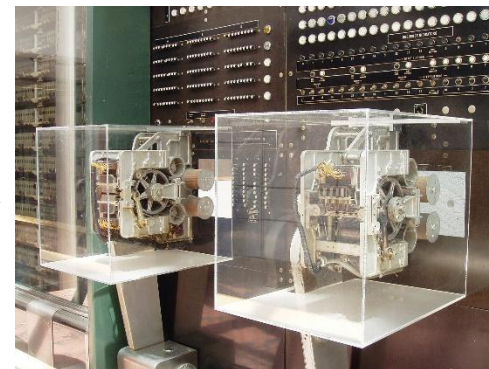
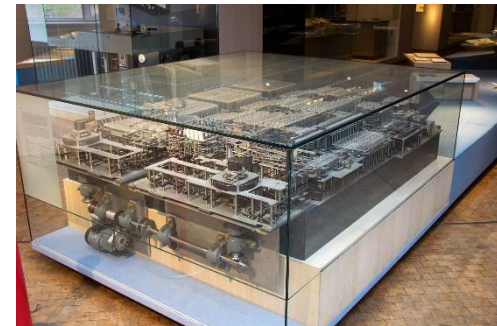
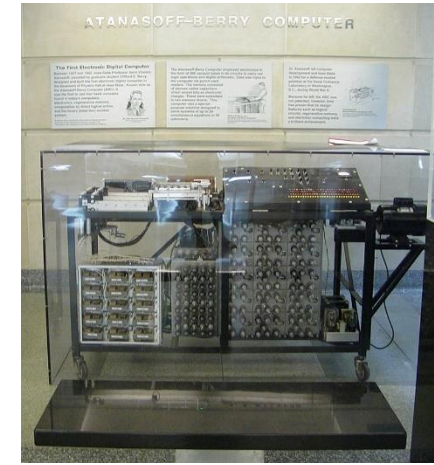
Between 1930 and 1950, several computers were invented by scientists who could be considered the pioneers of the electronic computer industry. 這些電腦代表了從機械計算到電子計算的重要過渡期，為現代儲存程式概念的發展奠定了基礎。

Early electronic computers

The early computers of this period **did not store the program in memory** — all were programmed externally.

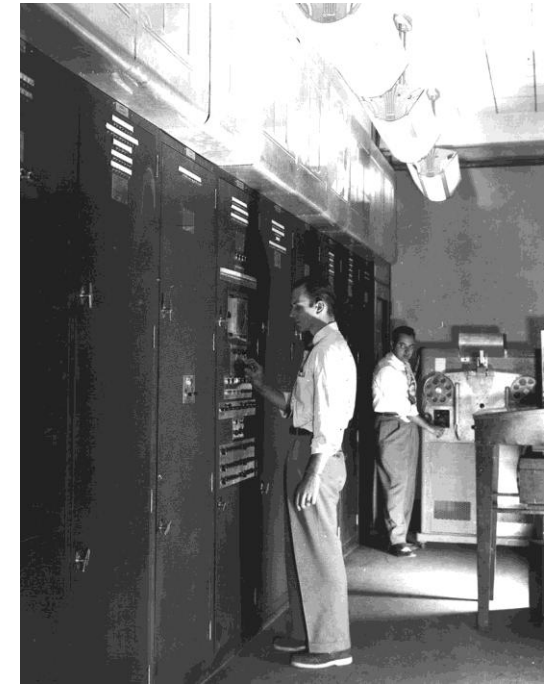
Five computers were prominent during these years:

- ABC (Atanasoff–Berry computer) 專門用於解線性方程組
- Z1 使用二進位系統
- Mark1 哈佛大學與IBM合作，大型電機械計算機
- Colossus 用於破解德軍密碼的電子計算機，在二戰中發揮關鍵作用
- ENIAC (Electronic Numerical Integrator and Computer)
第一台通用電子數位計算機，重達30噸，佔地167平方公尺



Computers based on the von Neumann model

- The preceding five computers used memory only for storing data, and were programmed externally using wires or switches.
- John von Neumann proposed that the program and the data should be stored in memory.
 - That way, every time we use a computer to do a new task, we need only change the program instead of rewiring the machine or turning hundreds of switches on and off.
- The first computer based on von Neumann's ideas was made in 1950 at the University of Pennsylvania and was called **EDVAC** (Electronic Discrete Variable Automatic Computer).
 - 第一台基於馮·紐曼概念設計的電腦
 - 採用了儲存程式概念 (stored-program concept) , 程式和資料都儲存在同一個記憶體中
- At the same time, a similar computer called **EDSAC** was built by Maurice Wilkes at Cambridge University in England.



The EDVAC as installed in Building 328 at the [Ballistic Research Laboratory](#).

1.4.3 Computer generations (1950–present)

- Computers built after 1950 more or less follow the von Neumann model. They have become faster, smaller, and cheaper, but the principle is almost the same.
- Historians divide this period into generations, with each generation witnessing some major change in hardware or software (but not in the model).

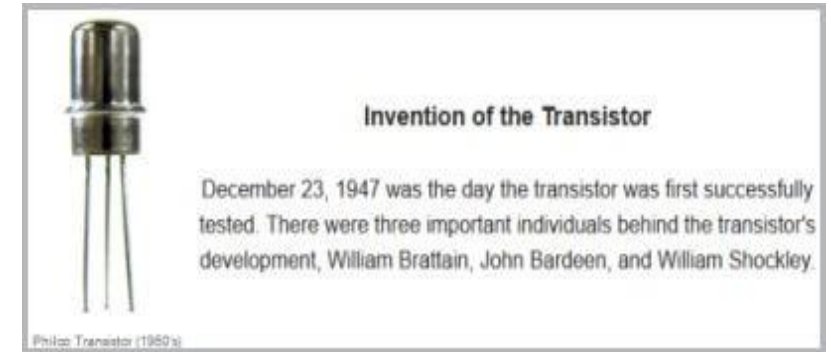
First generation: Vacuum Tubes 真空管

- The first generation (roughly 1950–1959) is characterized by the emergence of commercial computers.
- The UNIVAC (Universal Automatic Computer) and ENIAC (Electronic Numerical Integrator and Computer) computers are examples of first-generation computing devices.
 - The UNIVAC was the first commercial computer delivered to a business client, the U.S. Census Bureau in 1951.
 - UNIVAC的成功交付標誌著電腦產業從實驗室走向市場的關鍵轉捩點，為後來的商業電腦發展奠定了基礎。



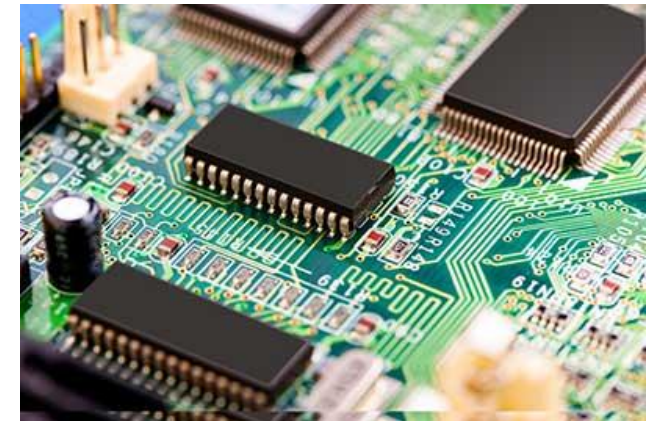
Second generation: Transistors 電晶體

- Second-generation computers (roughly 1959–1965) used **transistors** instead of vacuum tubes.
- [IBM 604](#) consists of 2,000 transistors. Clock speed of 58 kHz.
- Two high-level programming languages, FORTRAN and COBOL invented and made programming easier.



Third generation: Integrated Circuits 積體電路

- The invention of the **integrated circuit** reduced the cost and size of computers even further.
- Minicomputers appeared on the market.
- Canned programs, popularly known as **software packages**, became available.
- This generation lasted roughly from 1965 to 1975.
- [1964 IBM System/360](#)

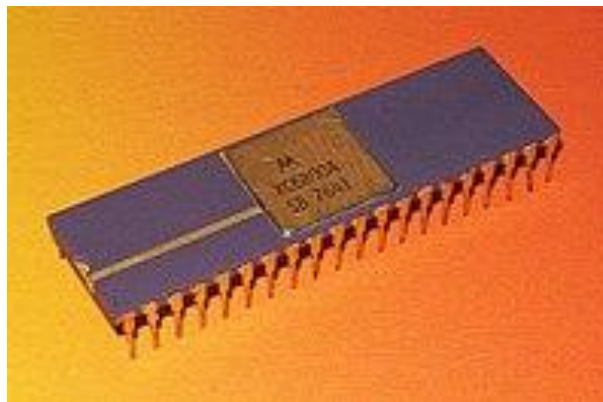


Fourth generation: VLSI Microprocessor Based 超大型積體電路

- The fourth generation (approximately 1975–1985) saw the appearance of microcomputers. [1974 Intel introduces the 8080 microprocessor](#)
- The first desktop calculator, the Altair 8800, became available in 1975.
- This generation also saw the emergence of **computer networks**.



[Intel 4004](#)



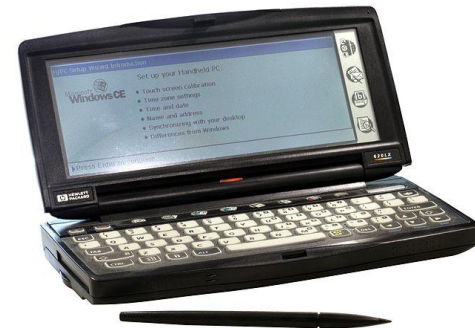
[Motorola 68000](#)
(MC68000)



A modern [64 bit x86](#) processor (AMD Ryzen 5 2600, Based on Zen+, 2017)

Fifth generation: Present and Beyond

- Fifth Generation Computer Systems - 1982 by Japan's Ministry of International Trade and Industry (MITI) to create computers using massively parallel computing and logic programming. VLSI, SLSI.
- This open-ended generation (1985~) has witnessed the appearance of **laptop** and **palmtop** computers, improvements in secondary storage media (CDROM, DVD and so on), the use of multimedia, and the phenomenon of virtual reality.



1.5 SOCIAL AND ETHICAL ISSUES 社會與倫理議題

Computer science has created some peripheral issues:

- Social issues
 - Dependency – a society in which the majority of people have access to a computer is considered a better society?
 - Social justice – all people can afford?
 - Digital divide – two groups: those who are electronically connected to the rest of the society and those who are not.
- Ethical issues
 - Privacy
 - Copyright
 - Computer crime

1.6 COMPUTER SCIENCE AS A DISCIPLINE

- With the invention of computers, a new discipline has evolved: **Computer Science**.
- Like any other discipline, computer science has now divided into several areas.
- We can divide these areas into two broad categories: systems areas and applications areas.
- This course is a breadth-first introduction to all these areas.
- After taking this course, students should have enough information to select their desired area of specialty.

1.7 OUTLINE OF THE COURSE

The book is divided into five parts.

1.7.1 Part I: Data representation and operation

- Chapter 2 discusses number systems, how a quantity can be represented using symbols.
- Chapter 3 discusses how different data is stored inside the computer.
- Chapter 4 discusses some primitive operations on bits.

1.7.2 Part II: Computer hardware

- Chapter 5 gives a general idea of computer hardware, discussing different computer organizations.
- Chapter 6 shows how individual computers are connected to make computer networks, and internetworks (internets).

1.7.3 Part III: Computer software

- Chapter 7 discusses operating systems.
- Chapter 8 shows how problem solving is reduced to writing an algorithm for the problem.
- Chapter 9 discusses efficiency of algorithms.
- Chapter 10 takes a journey through the list of contemporary programming languages.
- Chapter 11 is a review of software engineering.

1.7.4 Part IV: Data organization and abstraction

- Chapter 12 discusses data structure, collecting data of the same or different type under one category.
- Chapter 13 discusses abstract data types.
- Chapter 14 shows how different file structures can be used for different purposes.
- Chapter 15 discusses databases.

1.7.5 Part V: Advanced topics

- Chapter 16 discusses data compression.
- Chapter 17 explores some issues to do with security.
- Chapter 18 discusses the theory of computation.
- Chapter 19 introduces Computer Graphics.
- Chapter 20 gives some idea of artificial intelligence, a topic with day-to-day challenges in computer science.

1.7.6 Part VI: Social media and social Issues

- Part VI briefly discusses social media and social issues, two topics that students of computer science may be interested to explore.

Exercises

- P.15 1.9.2 Review questions
 - Q1-1 to Q1-10

Extend Readings

- The Imitation Game (電影:模仿遊戲)
- Tutorialpoint 電腦基礎介紹
- History of Computers - A brief Timeline (電腦發展史)